

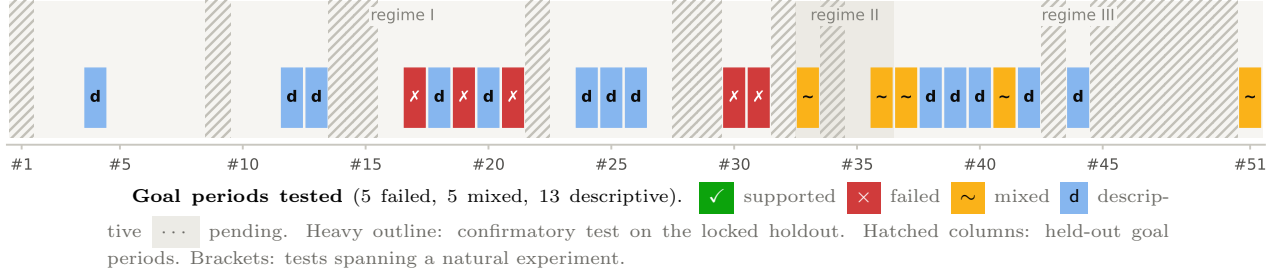
H133: Read-out Glauber Potts: an agent switches project only at a call, driven by the named messages it just read

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 45% Faithfulness 1/5 Usefulness 1/5

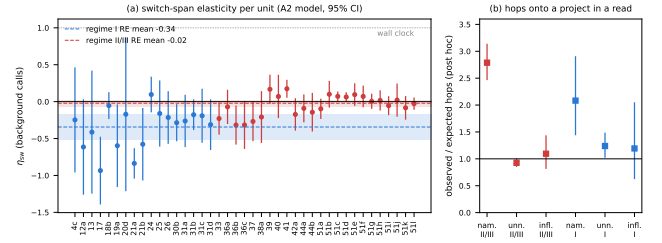
Bottom line. In regime II/III an agent's background project-switch hazard runs on its own call clock (switch-span elasticity $-0.02 [-0.07, +0.02]$, 24 units). Named-read coupling is untestable at the planted effect; post hoc, hops onto a project named to the agent are $2.8\times$ expected, unnamed $0.93\times$, in flight $1.1\times$.

level: hypothesis · pipeline checks 86% · updated 2026-10-07



WHAT HAPPENED

HH376 says that an agent changes project only at its own model call, by a Glauber update whose odds for project b rise with the read messages that name the agent and link b ; unnamed links add nothing, and without reads the switch rate is constant per call, not per hour. H133 labels every call with a project (shared `project_calls`: 2.6M calls, 61,939 non-reserved hops) and fits a conditional logit with project $\times$ active-hour effects in 41 non-reserved units. **The coupling test is untestable as planned:** named reads about a non-current project are so rare that a planted $\gamma_{\text{nam}} = 1$ gives under 2 such hops per unit (power ≈ 0 ; Amendment A1). **The clock test holds in regime II/III:** background switches do not scale with wall time ($\eta_{\text{sw}} = -0.02 [-0.07, +0.02]$, 24 units). The reserved periods are not run.



Observables. (a) Background switch-span elasticity η_{sw} per unit (A2 model, agent-cluster 95% CIs; blue regime I, red regime II/III; bands: random-effects means). No unit approaches the wall clock at 1. Regime II/III sits at 0; regime I is negative. (b) Post hoc, descriptive: hops onto a project in a read message, observed over the count expected without read terms, summed over units (Poisson 95%). Named reads: 2.79; unnamed: 0.93; named in flight (not yet readable): 1.10.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: an agent's spontaneous project switching is a per-call rate, so slowing its call rate slows its drift between projects, and long pauses do not add switches. Descriptively, a message that names an agent and links a project goes with about 2.8 times the expected switches onto that project. A link the agent reads without being named goes with none. To move an agent, name it in the message with the link. This rule is not yet a tested result.

$$u_{\text{stay}} = \alpha_i + \phi \ln(1 + d)$$

$$u_b = \kappa_{b,h} + \sum_k \gamma_k \ln(1 + N_b^k) + \rho[\text{held}] + \beta_s s_b$$

$$\text{cloglog } P(\text{hop} \mid \text{no read}) = \alpha_i + \eta_{\text{sw}} \ln \text{span}_c + \dots$$

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RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
F: O1–O3 testable	power ≈ 0 per unit at $\gamma = 1$	A1
P1 $\gamma_{\text{nam}} > 0$	9/24 units (13 estimable); RE +2.3	untestable
P2 $\gamma_{\text{un}} = 0, \Delta\gamma > 0$	RE $\gamma_{\text{un}} + 0.09 [-0.16, 0.34]$	untestable
P3 named $>$ in flight	γ_{if} in 3 units	untestable
P4 $\eta_{\text{sw}} \ni 0, \not\ni 1$ (II/III)	$-0.02 [-0.07, +0.02]$	holds
P5 regime I $\eta_{\text{sw}} > 0$	$-0.34 [-0.51, -0.17]$	failed
P6 read-hop lag ≤ 1 : ≥ 0.55		not met
0.6		
N2 #51 timer wakes $\eta \ni 0$	$+0.46 [+0.09, +0.84]$	failed
N4 G31 $\eta > 0$	$-0.22 [-0.34, -0.09]$	failed
Kill B (wall clock)	CI excludes 1 in 41/41	not fired

GOAL PERIODS IN DETAIL

G04 d descriptive
G12 d descriptive
G13 d descriptive
G17 $\times$ failed
G18 d descriptive
G19 $\times$ failed
G20 d descriptive
G21 $\times$ failed
G24 d descriptive
G25 d descriptive
+13 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

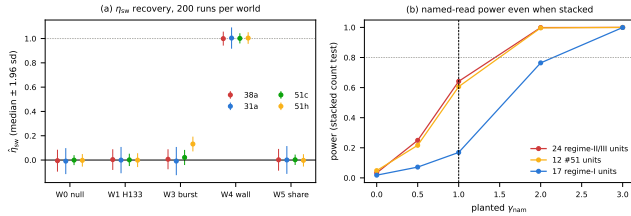
The read ratios are post hoc and descriptive: they were added after the synthetic run made the planned test untestable, and units with few named hops are noisy. A named message may answer the reader's own earlier mention of the project (not yet controlled). Call hops are fine-grained: 26% return to the previous project within 5 calls. $\eta_{\text{sw}} < 0$ in regime I and > 0 at timer wakes are unexplained; a burst field can also give $\eta \approx +0.1$.

ROUND 2

Pre-register the named-read ratio with power at $\gamma \approx 2$ and hierarchical shrinkage; control for the reader's own last touch of the project; split spans by gap kind to explain the regime-I and timer-wake elasticities. Confirm C1 (call clock) on #45–#47 and the #51 tail.

- **Essence.** whether naming an agent in a message that links a project moves that agent onto the project, beyond the attention burst that produces the message. Round 1 saw a $2.8\times$ lift only post hoc, so round 2 must test it with power at the observed scale.
- **H133-R1** Test the named-read ratio with power at $\gamma \approx 2$.
- **H133-R2** Add a sender-side impostor.
- **H133-R3** Explain $\eta \ni 0$ in regime I and $\eta \not\ni 0$ at timer wakes.
- **H133-R4** Fit the deferred variants.

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Second figure. Synthetic validation on four real skeletons. (a) η_{sw} is recovered without bias in null, H133, share and wall-clock worlds (200 runs each); an attention burst biases it to $+0.13$ on 51h. (b) Even stacking all 24 regime-II/III units, a count test of named-read hops has power 0.64 at the planted $\gamma_{\text{nam}} = 1$. It reaches 1.0 only at $\gamma_{\text{nam}} = 2$, so the per-unit coupling test was declared untestable before real data.